
Software Requirements Specification

for

HomeZapp

Version 1.0 approved

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Group 6

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Revision History

Date	Version	Description	Authors
1/27/2026	SRS Version 1	Initial SRS document	Pragati Gautam, Dean Patterson, Zaccheus Little, Dalton Washington
4/6/2026	Version 2	Full revision, fixed duplicate requirements from version 1, added section 5	Pragati Gautam, Dean Patterson, Zachheus Little, Dalton Washington

1. Introduction

HomeZapp is a platform that allows customers to either buy or sell real estate properties. These properties are made available for the buyers to view as listings and categorized into the 6 major architectural aesthetics- Colonial, Victorian, Farm, Mediterranean, Craftsmen, Modern, and other. The users are divided into 3 major classes- buyers, administration, and sellers. The website is built on Django stack, all the real estate property information and all other saved information is saved into a SQLite database, and the project uses HTML/CSS for frontend

1.1 Purpose

The software requirements specification document is intended to provide a detailed report of the features, design, functionality and constraints of the product. The document also serves the purpose of communicating the software development team's planning to the board members and investors of HomeZapp. Additionally, this document can also be used as a point of reference for future employees and developers in case they want to make any modifications to the ecommerce platform. This version of the SRS (Version 2.0) expands on the feedback received from version 1.0 and adds more requirements and user stories.

1.2 Document Conventions

This document uses Times New Roman font throughout the entire document. Major header lines use bold, size 24 font. Subsections use bold, italics, size 18 font. Smaller subsections use bold, italic, size 13 font. Any part of this document that is not a header uses size 11 font.

1.3 Intended Audience and Reading

- Project Manager(s)
- Software Developers
- Testers
- Investors
- Stakeholders

1.4 Product Scope

This is a website aiming to provide categorization and useful information to potential house buyers. It will benefit buyers because they can see relevant information about the homes and private sellers as the website will require them to give necessary information to make sure they are marketing themselves well. Our goal is to provide a complete process for home buyers and sellers trying to find/sell their home. Our objectives are to provide guidance to inexperienced home sellers and options for buyers. Our business goal is to make as many transactions as possible between buyers and sellers by matching customers with their desired house and providing important information about the house and seller, as well as information to the seller about potential buyers.

1.5 References and Acknowledgements

- Zillow for website layout and functional reference- [Zillow: Real Estate, Apartments, Mortgages & Home Values](#)
- Django- [The web framework for perfectionists with deadlines | Django](#)
- Python- [Welcome to Python.org](#)
- MySQL- <https://www.mysql.com/>

2. Overall Description

2.1 Product Perspective

HomeZapp is a new, self-contained digital project with functions designed to connect the buyer directly to the seller under the management of admins. There is no initial framework or system to build upon outside of Django; this software must be built in two parts, with a frontend and backend. The frontend will allow the user to interact with the webpage using a built-in interface. In the backend, our software will allow us to store and aggregate data. The web application itself will be built on top of Django.

As stated by the customer, this product must have users with different roles, the buyer should be able to search, compare, cancel the process for, and buy different types of products, a seller should be able to add, sell, and receive payments for their products; and an admin should be able to approve/block new user accounts and products.

2.2 Product Functions

The product -

- Connects buyers to sellers and vice versa.
- Make sure that the buyer has all information needed to make a decision about the house.
- Allow reports to be made by users about listings
- Makes it so the buyer or the seller can pull out of the house application/purchasing process
- Handles the actual payment from buyer to seller
- Lets users browse houses based on architectural features and not just prices
- Provides educational information about the different architectural types to help users make an informed decision

2.3 User Classes and Characteristics

The main expected user classes for this project are home buyers, home sellers, and administration. Home buyers will be identifiable internally through their email, which can be used to block malicious accounts by administration. Home sellers can post, edit, and delete their listings. Once a buyer has found a listing, they can apply for the listing and negotiate the agreed upon price. They also have a set amount of time to pay the price. Administrative users can remove listings and block buyer and seller accounts. The buyer and seller users are the most important.

2.4 Operating Environment

Backend: We chose Django, since it is a Python framework, a language that we are already familiar with and have great experience in. This makes our application easier to maintain and further add stuff if and when needed.

Database: We chose MySQL for the database because it is most readily integrated in Django. Since MySQL is good at handling relations efficiently, it is well suited for our website.

Frontend- We chose HTML/CSS since we are already familiar with those and also to keep our frontend simple and manageable.

2.5 Design and Implementation Constraints

Since we are mostly accommodating to American users, the website is only available in English, so we will not be able to accommodate people who are more comfortable in other languages. The verification software for sellers is not included in the website program.

3. System Features

3.1 Create Account

3.1.1 Description and Priority

Priority: High

New users who navigate to the website can create an account to be stored within our database. This allows them to access buyer features or be verified to access seller features.

3.1.2 Stimulus/Response Sequences

User stories

- User needs to create a buyer account using their email and password so that they can view, wishlist, apply for and buy houses.
- User needs to create a seller account using their email address and password so that they can publish a house listing, view potential buyer applications for their listing and receive money for selling their house listing. The user also needs to be notified when their seller verification gets approved by the admin.
- The user (seller) needs to be aware when their seller verification fails.

Sequence 1 (Buyer registration)

- 1) User navigates to the website
- 2) User selects “Create Account” on the homepage
- 3) System redirects user to create account page
- 4) User inputs required information, selects buyer account type, & submits form
- 5) System creates account and stores within its database
- 6) The system logs the user in and redirects them to the buyer user dashboard.

Sequence 2 (Seller account with verification successful)

- 1) User navigates to the website
- 2) User selects “Create Account” on the homepage
- 3) System redirects user to create account page
- 4) User inputs required information, selects seller account type, & submits form
- 5) Admin manually approves the account
- 6) Admin alerts system verification was successful
- 7) System creates seller account and stores within its database
- 8) System redirects the user from the create account page to the seller user dashboard.

Sequence 3 (Seller account with verification unsuccessful)

- 1) User navigates to the website
- 2) User selects “Create Account” on the homepage
- 3) System redirects user to create account page
- 4) User inputs required information, selects seller account type, & submits form
- 5) Admin manually declines the request
- 6) Admin alerts the user that system verification was not successful
- 7) The user is notified on the create account page the verification process was not successful.

Sequence 4

- 1) User navigates to the website

- 2) User selects “Create Account” on the homepage
- 3) User inputs required information incorrectly, selects either seller or buyer account type, & submits form with incomplete or invalid information
- 4) The user is notified of incorrect credentials

3.1.3 Functional Requirements

REQ-1: The system should let users register using their email ID, full name and account type

REQ-2: The system should reject registration in case of an email ID already being associated with an existing account.

REQ-3: The system shall redirect all seller accounts to the admin notifications tab for the admin to accept or reject

REQ-4: The system shall store all passwords in an encrypted (hash) form

REQ-5: The system should communicate with the admin

3.2 Login

3.2.1 Description and Priority

Priority: Medium

The users will be prompted to login to their account using their email and password. On a successful login attempt, the user will be signed into their account and redirected to their buyer/seller homepage.

3.2.2 Stimulus/Response Sequences

User Stories-

- As a buyer user class, I should be able to login into my account to view saved listings , wishlists, complete the buyer application form and be able to pay for a house.
- As a seller user class, I should be able to login into my account and be redirected to the seller dashboard and be able to manage my listings, view application forms for buyers who are interested in the property and be able to negotiate prices with a potential buyer.

Sequence 1- Successful buyer login

- 1) The user selects the login option from the HomeZapp homepage
- 2) The system redirects the user to the login page
- 3) The user enters their email and password associated with their account
- 4) The system verifies the email and password as correct, signs the user in to their account, sees that they are a buyer, and redirects them to the homepage

Sequence 2- Failed user login

- 1) The user inputs incorrect email/password
- 2) The system notifies the user of this and does not sign them in

Sequence 3- Successful seller login

- 1) The user inputs correct email/password
- 2) The system verifies the login as correct, logs them into their accounts, sees the user is a seller, and redirects them to their seller dashboard

3.2.3 Functional Requirements

REQ-6: The system allows users to log in using an email address and password from the stored database

REQ-7: The system displays an error message if the user enters incorrect information into login

REQ-8: The system sends users to the correct dashboard type tied to their user class(buyer/seller)

REQ-9: The system shall create and store a cookie whenever a user logs in to maintain the user's session until they log out of their account

3.3 Seller Verification

3.3.1 Description and Priority

Priority: Medium

Users intending to sell a house have their details sent to the admin for verification process before posting listings. This allows the platform to be more secure and reduce fraudulent activity.

3.3.2 Stimulus/Response Sequences

Sequence 1- Seller account approved

- 1) Whenever a seller creates an account, their credentials are sent to the admin
- 2) The admin receives a notification on their admin dashboard and reviews the request
- 3) The system shows the user that their account is under review
- 4) The admin then decides that they want to approve the seller account
- 5) The seller receives a message saying their request has been approved

Sequence 2- Seller account not approved

- 1)Whenever a seller creates an account, their credentials are sent to the admin
- 2) The admin receives a notification on their admin dashboard and reviews the request
- 3) The system shows the user that their account is under review
- 4) The admin then decides that they want to approve the seller account
- 5) The seller receives a message saying their request has been approved

3.3.3 Functional Requirements

REQ-10: The system should send seller credentials to the admin for the admin to review, accept or reject

REQ-11: The system should notify the seller user class when their account is being reviewed by admin

REQ-12: The system should notify the user class when their account has been approved by the admin

REQ-13: The system should notify the seller user class when their account has been rejected by the admin

REQ-14: The system should not allow users who are not sellers to access listing features until verification is approved by the admin class.

3.4 Update Account information

3.4.1 Description and Priority

Priority: Low

Users can edit their account information after they have logged in.

3.4.2 Stimulus/Response Sequences

User story- As a registered user, I should be able to edit my information when logged in to make my account updated and secure.

Sequence 1

- 1) User logs in to the account
- 2) User goes to manage account
- 3) The system redirects the user to manage account page
- 4) User selects "Edit profile" and submits information
- 5) The system validates the information and updates the data in the database and thus it reflects on the website too

3.4.3 Functional Requirements

REQ-15: The system should let users edit and update stored account information

REQ-16: The system should validate and save the data

REQ-17: The system shall allow only logged in users to be able to update their account information

REQ-18: The system shall verify updated email addresses for being unique (there shouldn't be 2 users with the same email address) and being in the correct format (eg- myname@gmail.com is correct but myname.com is incorrect)

3.5 Search for listings

3.5.1 Description and Priority

Priority: High

Users can search for listings based on their requirements and needs. This is a high priority feature and sits at an 8 out of 10, since without this feature the users would just be looking at random listings and wouldn't be able to narrow down to their requirements.

3.5.2 Stimulus/Response Sequences

User Stories-

- As a buyer I want to search for listings based on city and architectural styles
- As a buyer, I don't want to spend more than 5 seconds at the same page

Sequence: Buyer searching for a listing

- 1) User goes to homepage and clicks on search bar
- 2) User types in what kind of house they want in their city of preference and/or bed/bath requirements
- 3) The system filters the houses from the database and shows the user their preferences within less than 5 seconds
- 4) User can now view all of the options available

3.5.3 Functional Requirements

REQ-19: The system should allow users to search for criteria (Location, architectural style, price, etc.)

REQ-20: The system should be able to retrieve and display listings that match the criteria given by the user.

REQ-21: The system should be able to show results within less than 5 seconds of the user pressing the search button

REQ-22: The system shall only return active listings for the buyer to view.

REQ-23: The system shall provide a search bar at the buyer dashboard as well as the main homepage for the users to search for houses

3.6 Filter listings

3.6.1 Description and Priority

Users can sort and filter listings based on what they are looking for and filter based on prices, location proximity, and relevance. It is a low-priority task and is not essential for the website to work, but it would improve the user experience. Because of this, it is a medium priority.

3.6.2 Stimulus/Response Sequences

User Story- As a buyer, I want to be able to sort listings based on prices and filter them based on architectural styles

- 1) The user searches for an item on the search bar
- 2) The system gives the user the available houses that match the search
- 3) The user chooses the sort option that they want from the dragdown options (Price:low to high, Price:high to low, Newest)
- 4) The system sorts according to the user's preferences
- 5) The user applies filters like location, price, beds and baths, and architectural types
- 6) The user clicks on the search button after choosing all the filter and sort options
- 7) The system updates the website to show the intended listings to the user

3.6.3 Functional Requirements

REQ-24: The system should allow users to apply sorting and filtering options

REQ-25: The system should update and display the listings based on the user selected filters

REQ-26 : The system should return the search results within less than 5 seconds of pressing search button

REQ-27: The system shall display only active listings to the buyer

REQ-28 : The user should be able to clear all applied filters

3.7 Buyer Application Form Submission

3.8.1 Description and Priority

Priority: High

When the buyer is interested in a property, they can show interest to the seller by filling out the buyer application form and submitting it. This form includes legal documents and paperwork required for the seller's viewing.

3.8.2 Stimulus/Response Sequences

User story-

- As a buyer, I want to be able to submit a formal application to express my interest in a property and should be able to submit legal documents in the form of pdfs
- As a buyer, I want to be notified when my application has been submitted so that I know that my information has been sent to the seller.

Sequence 1

1. The buyer views a listing and decides that they are interested in it. The buyer fills out the Buyer application form and uploads their legal documents and paperwork in there including government issued ID, mortgage pre-approval documents, and bank statements. They then submit the form
2. The system saves the form and displays it in the listing's seller's dashboard
3. The systems shows buyer application form submitted on the user's dashboard

Sequence 2

1. The buyer enters their information and documents
2. The system recognizes documents are missing/bad information was given and alerts the buyer.

3.8.3 Functional Requirements

REQ-29: The system should allow buyers to enter required details into the buyer application form

REQ-30: The system should validate file uploads in PDF, JPG and PNG formats

REQ-31: The system should be able to recognize incomplete information and inform the user

REQ-32: The system should notify the buyer when their buyer application form has been submitted so that they know that their information has been sent to the seller.

REQ-33: The system shall only allow logged in buyers to submit a buyer application form for a particular listing that they are interested in

REQ-34: The system shall display submitted applicants on the seller's dashboard

REQ-35: The system shall notify the respective seller when a buyer application form has been received

REQ-36 : The system shall allow a buyer to submit applications for more than one house listing.

3.8 Accept/ Reject/ Negotiate Deal***3.8.1 Description and Priority***

Priority: High

3.8.2 Stimulus/Response Sequences

User stories-

- As a seller I want to accept a buyer's application so that they can make the payment
- As a seller, I want to be able to decline a buyer's application for a stated reason so that the buyer can know what reason they were declined for.
- As a buyer or seller, I want to be able to negotiate with the other party to get a better price for myself

Sequence 1- (Accept)

1. A buyer receives a notification from the seller saying their buyer application has been accepted
2. The buyer receives a notification on their notifications tab that when clicked on, takes them to the payment page. The buyer then fills out their payment details and the transaction is completed

Sequence 2- (Reject)

1. The buyer submits the buyer application form
2. The buyer receives a notification saying their application has been rejected for a reason that'd be stated by the seller
3. The buyer gets to apply one more time if it's an issue with their documents

Sequence 3- (Negotiate)

1. The buyer submit the buyer application form for the listing that they are interested in
2. The buyer receives a notification saying their application has been accepted for that listing
3. The buyer sees the notification on their notifications tab, but instead of going to the payment page, they click on the negotiate button and type in their desired price value
4. The negotiation goes forward to the seller- and they make a decision whether to accept or reject deal

3.8.3 Functional Requirements

REQ-37: The buyer should be able to negotiate with the seller in case they would want to

REQ-38: The system shall allow sellers to accept or reject a buyer application and the application is treated as a contract between the two parties

REQ-39: The system shall notify the buyer of their application being rejected and the seller's stated reason for doing so

REQ-40: The system shall notify the buyer of their application being submitted and then, redirect the buyer to the payment page

3.9 Payment Page

3.9.1 Description and Priority

Priority: High

After negotiations, buyers will be navigated to a payment page to submit payment. At this step, both parties are unable to back out of the selling process

3.9.2 Stimulus/Response Sequences

User Story-

- ***As a buyer, I should be able to enter my bank account details to make a payment on the house.***

Sequence 1

- 1) After negotiations, the buyer chooses to navigate to “Pay and Sign.”
- 2) The system redirects user to Payment Page
- 3) User completes the process by uploading their payment information
- 4) The system processes the transaction from the buyer to the seller and sends notifications to both that the payment has been submitted and the process is finished

Sequence 2

- 1) The buyer inputs incomplete payment information
- 2) The system informs the buyer of the incomplete information

3.9.3 Functional Requirements

REQ-41: The system should be able to handle transactions and make them secure.

REQ-42: The system should be able to send the notifications to buyers when their payment has been received by seller

REQ-43: The system shall never store any payment information on the database

REQ-44: The system shall change the listing status to sold when a buyer completes the payment

REQ-45: The system shall record a payment confirmation with seller ID, buyer ID, listing ID and amount paid along with the timestamp

REQ-46: The system should be able to recognize incomplete payment information and inform the buyer

3.10 Ban User

3.10.1 Description and Priority

Priority: Medium

Admin is able to ban a user if the user is found to be fraudulent and/ or if they violate HomeZapp’s policy agreement.

3.10.2 Stimulus/Response Sequences

User story-

- As an admin, I am able to ban a user if they do any fraudulent activity or if they break any terms of the policy contract for HomeZapp.

Sequence: Admin banning a user

- 1) A user has submitted a complaint about another user
- 2) The admin views the complaint and decides to ban the user based on the context
- 3) The system records that the user is banned and will not allow the user back into their account. Attempts to make a new account with the same email are also prevented by the system.
- 4) The system also notifies the user with the email address they had registered with HomeZapp that their account has been banned and states the reason for doing so as well.

3.10.3 Functional Requirements

REQ-47: The system should allow admin users to ban user accounts(both buyer or seller) that violate HomeZapp policies

REQ-48: The system should disable the banned user's access to the platform

REQ-49: The system should be able to prevent attempts to create new accounts with the same email

REQ-50: The system shall log all ban actions including timestamp, userID, adminID, and reason for ban

REQ-51: The system shall automatically remove all active listings of a banned seller

REQ-52: The system shall notify of a buyer being banned to the sellers they were in negotiation process with

REQ-53: The system shall require the admin to state a reason for ban

REQ-54: The system shall notify the user on their email ID about the admin banning and give them the reason for doing so

3.11 Send notifications to buyers or sellers

3.11.1 Description and Priority

Admin is able to send alerts or notifications to the buyers and sellers through the website. It is a medium-priority task since it's not required for the website to work but it's also not a frivolous feature.

Priority: Medium

3.11.2 Stimulus/Response Sequences

User story:

- As an admin, I should be able to send platform-wide notifications or targeted notifications to both buyer and seller user classes
- 1.) The website has a notifications tab that takes the users to a notifications inbox
 - 2.) The notifications can vary from- new features to the website, update in terms of agreement and alerting a user of their interaction with a fraud user
 - 3.) Whenever the users login, they can view their personalized notifications on the notifications inbox

3.11.3 Functional Requirements

REQ-55: The system should allow admins to create and send notifications to all user classes

REQ-56: The system should store notifications and display them in the user's notification box.

3.12 Send Complaints

3.12.1 Description and Priority

Priority: High

This feature allows users to submit complaints about suspicious behavior. Although sellers will be verified before being able to sell, there could still be a chance of a bad actor. This gives users the ability to help us catch those bad actors, while also giving sellers the ability to tell us about buyers that seem shady. Complaints will be sent to administrative users.

3.12.2 Stimulus/Response Sequences

User Story-

- As a user, I should be able to submit complaints to report another user.
- 1) A signed in user selects the "Report" option
 - 2) The user submits their complaint form
 - 3) The system then records the complaint and sends it to every admin user's notification box

3.12.3 Functional Requirements

REQ-57: The system should allow any user to submit complaints regarding any user or listing

REQ-58: The system shall require the complaint to have a written description

REQ-59: The system shall record the complaint in the database with timestamp, user ID- of both the user reporting and the user being reported

REQ-60: The system shall deliver complaint notifications to all admin inboxes

3.13 Wishlist listings

3.13.1 Description and Priority

Priority: High

This feature allows users to wishlist property listings for future reference. Saved listings will help users easily revisit a home they were interested in without needing to search again, which improves the general flow and feel of the website.

3.13.2 Stimulus/Response Sequences

User stories-

- As a buyer, I want to be able to wishlist a listing to save it for future
- As a buyer, I should be able to remove a listing from my wishlist if I want

Sequence One: Add listing to wishlist

- 1) The user will sign in to their account
- 2) The system validates information and signs the user in, then redirects to the homepage
- 3) The user will browse the listings, and then select the save listing option on something they're interested in
- 4) The system records the listing in that buyer's saved listings collection
- 5) The user can then access their saved listings later.

Sequence Two: Remove listing from wishlist

- 1) The user will sign in to their account
- 2) The system validates information and signs the user in, then redirects to the homepage
- 3) The user will browse the listings, and then select the remove from wishlist option on something they're not interested in anymore
- 4) The system removes that listing from that buyer's saved listings collection

3.13.3 Functional Requirements

REQ-61: The system should allow a signed-in user to wishlist any property listings

REQ-62: The system should be able to store saved listings in the user's account.

REQ-63: The system shall store the association between the buyer's account and the listing in the database

REQ-64: The system shall provide a 'my wishlist' page for all buyers to view their wishlisted properties

REQ-65: The system shall prevent a user from wishlisting an already wishlisted listing

3.14 Create listings

3.14.1 Description and Priority

Priority: High

This feature will allow sellers to create a property listing on the website. Sellers must provide required details about the property, such as location, description, price, images, and architectural style. The seller's ability to create listings is important for connecting buyers and giving buyers something to browse on the website.

3.14.2 Stimulus/Response Sequences

User Stories-

- As a verified seller, I want to be able to create a listing using images, physical address, architectural type, and price so that potential buyers can view my property.

Sequence One: Successfully created listing

- 1) The seller logs into their seller-verified account
- 2) The system validates the information and redirects user to homepage
- 3) The seller navigates to the "Create Listing" tab
- 4) The system redirects user to "Create Listing" tab
- 5) The seller enters property details (address, price, description, images, architectural style) and submits the information
- 6) The system stores the listing in the database and sends the information to admin
- 7) The admin then decides to approve it

Sequence Two: Listing already exists

- 1) The seller logs into their seller-verified account
- 2) The system validates the information and redirects seller to homepage
- 3) The seller enters the property details of an already existing listing
- 4) The system informs the seller of Error: Listing already exists

3.14.3 Functional Requirements

REQ-66: The system allows sellers to create property listings

REQ-67: The system should require sellers to provide listing details before being able to publish anything

REQ-68: The system should prevent the seller from publishing duplicate entries

REQ-69: The system should send a 'pending approval' message when a listing is submitted by a seller

REQ-70: The system should send a 'listing approved' message when a seller's listing has been approved by admin

REQ-71: The system should send a 'listing not approved' message when a seller's listing has been denied by admin

REQ-72: The seller user class shouldn't be able to create a listing and make it show on the website without approval from the admin user class

REQ-73: The system should accept uploads in jpg, png or jpeg formats for the property's pictures.

3.15 Manage listings

3.15.1 Description and Priority

Priority: High

This feature allows sellers to manage their listings on the website. Sellers will be able to edit, add, or remove listings as they see fit, ensuring accurate and up-to-date information.

3.15.2 Stimulus/Response Sequences

User stories-

- As a verified seller, I am able to edit my listing information to keep the property information accurate and upto date
- As a verified seller, I am able to delete any listings that I have published if I need to take my property off the market

Sequence One: Seller is able to edit a listing information

- 1) The seller logs into their verified account
- 2) The system verifies information and redirects them to homepage
- 3) The seller navigates to their listings page
- 4) The system redirects the seller to listings page
- 5) The seller selects edit listing
- 6) The system redirects seller to edit listing page
- 7) The seller updates the listing information and submits the change
- 8) The system validates the saves and updates the listing

Sequence Two: No changes

- 1) The seller logs into their verified account
- 2) The system verifies information and redirects them to homepage
- 3) The seller navigates to their listings page

- 4) The system redirects the seller to listings page
- 5) The seller updates the listing information with information already present on the listing and submits change
- 6) The system informs the seller of Error: The information entered is already present in database

3.15.3 Functional Requirements

REQ-74: The system should allow users that are signed in to edit the information of listings they have created

REQ-75: The system should allow sellers to remove their own listings.

REQ-76: If the seller doesn't make any change to the listing and hits submit, the system shall say-information already exists

REQ-77: The system shall prevent sellers from editing listings which are created by other sellers

REQ-78: If a seller removes a listing, it shouldn't be visible to buyers.

3.16 Log Out

3.16.1 Description and Priority

Priority: High

This feature allows all users to log out of their account, to invalidate their active session

3.16.2 Stimulus/Response Sequences

User story- As a user, I want to be able to log out whenever I want and that should end my access to my account until I login back again

- 1) Any user class(buyer/seller/admin) clicks log out
- 2) The system ends their session cookie
- 3) The user loses access to their saved information on the website until they login back again

3.16.3 Functional Requirements

REQ-79: The system can make it so users have to sign in again before they can access their account

REQ-80: The system should provide all users with a logout option from their account page

REQ-81: Upon logging out, the system ends the user session's cookie so that loading pages again require re-authentication

REQ-82: The system shall automatically invalidate a session if it has been idle for more than 30 minutes

3.17 Admin Login

3.17.1 Description and Priority

Priority: High

This feature enables admins to log into their account to access their admin dashboard

3.17.2 Stimulus/Response Sequences

User story-

As an administrator, I want to be able to login into my admin account to have access for more than a user account

Sequence One: Successful Admin Login

- 1) An admin user on the admin login page inputs their information correctly
- 2) System logs them into their account and redirects them to the admin dashboard

Sequence Two: Failed Admin Login

- 1) An admin user on the admin login page inputs their information incorrectly
- 2) The system notifies the user of this

3.17.3 Functional Requirements

REQ-83: The system should log users in and store a cookie so that they are recognized until they log out and the cookie is deleted

REQ-84: The system shall notify users if their information is incorrect

REQ-85: The system shall have a separate admin login page- different from other users login

3.18 Admin Create Account

3.18.1 Description and Priority

Priority: High

Admin can create an account to be stored within our database. This allows them to access admin options such as ban users, delete listings and send notifications to the user class.

3.18.2 Stimulus/Response Sequences

Sequence 1

- 1) When the admin class types website url/ admin they are taken to the admin login page
- 2) On the admin login page, the admin class can click on the 'admin create account' option
- 3) The system redirects the user to the admin create account page
- 4) After getting redirected to the admin create account page, the admin can enter their email ID and if it's verified with the system, they will get a verification code on their email ID
- 5) If their email ID is a part of the system, they are able to create a password for their email ID, after which they will get redirected to the admin login page

Sequence 2

- 1) When the admin class types website url/ admin they are taken to the admin login page
- 2) On the admin login page, the admin class can click on the 'admin create account' option
- 3) The system redirects the user to the admin create account page
- 4) After getting redirected to the admin create account page, the admin can enter their email ID and if it's verified with the system, they will get a verification code on their email ID
- 5) If their email ID is not a part of the system, the website will say email is not verified.

3.18.3 Functional Requirements

REQ-86: The system shall have a list of emails that are the verified admins so that those admins can create an account and login

REQ-87: The system shall tell the admin if their email is not verified yet and thus they wouldn't be able to create account

3.19 Admin Remove Listing**3.19.1 Description and Priority**

Priority: High

This feature enables admins to remove any listings from the website- especially in case of fraud activity, use of abusive language or any other HomeZapp policy violations.

3.19.2 Stimulus/Response Sequences**User story-**

- As an admin user class, I should be able to remove a listing based on fraud activity or HomeZapp policies violation
- As an admin user class, if I try to remove a listing that has already been removed, the system should show an error.

Sequence One: Admin removes listing

- 1) Admin selects a listing that it has received complaints against and decided to remove it from the admin dashboard
- 2) Admin provides a reason to remove and confirms removal
- 3) System sends a notification to the seller who posted the listing along with reason for removal and sets the listing to admin removed on their seller dashboard

Sequence Two: Admin is unable to remove listing

- 1) An admin tries to remove a listing that has already been removed
- 2) The system notifies the admin that the listing has already been removed

3.19.3 Functional Requirements

REQ-88: Admin should be able to remove any listing regardless of any factors

REQ-89: The system shall log every action related to removing listing ie- Admin ID, timestamp, reason for removal etc

REQ-90: The system shall display an error when an admin tries to remove a listing that is already removed

REQ-91: The system shall notify the seller when their listing gets removed by admin and also give them the reason why

3.19 Admin Approve/ Deny Listing

3.19.1 Description and Priority

Priority: High

This feature enables admins to approve or deny listings posted by the seller to make sure that every listing posted on the website is admin approved

3.19.2 Stimulus/Response Sequences

User story-

- As an admin user class, I should be able to approve a listing put up by a seller
- As an admin user class, I should be able to deny a listing put up by a seller based on factors like incomplete information, fraud detection or any HomeZapp policy violation

Sequence One: Admin approves a listing

- 1) Admin receives a notification on their dashboard saying that a seller has submitted a listing information
- 2) Admin reviews listing information and confirms it
- 3) Listing gets posted to the website and is available for buyers to view

Sequence Two: Admin denies a listing

- 1) Admin receives a notification on their dashboard saying that a seller has submitted a listing information
- 2) Admin reviews listing information and denies it and states a reason for doing so

- 3) The seller is notified that their listing has been denied and they receive a reason for the same

3.19.3 Functional Requirements

REQ-92: The system shall display all listings with 'pending approval status' on the admin dashboard

REQ-93: The system shall display all of the listing details with the listing on the admin dashboard

REQ-94: The system shall show the date and time the listing was submitted for approval

REQ-95: The system shall prevent buyers from viewing a listing before it has been approved

REQ-96: The admin is able to reject a listing if they suspect any fraud, abusive language or policy violations

REQ-97: The seller should be notified when their listing is accepted

REQ-98: In case of a listing rejection, the seller should be notified and the reason for rejection should be stated.

REQ-99: The admin shouldn't be able to accept/ reject a listing that isn't in the 'pending approval status' and has already been either approved or rejected.

REQ-100: The time and date that action was taken should be saved with the record of the approval/denial of the listing

4. Other Nonfunctional Requirements

4.1 Performance Requirements

- 1) Speed and Responsiveness- The home page, filter pages, and search pages must be loaded and interactive relatively fast
- 2) Capacity- the website is able to support 10,000 users simultaneously without quality degradation

4.2 Safety Requirements

- 1) Age verification- User must be at least 18 years old to be able to create an account and they would need to verify their age through their email
- 2) Hate speech removal- Any user that participates in hate speech can be flagged for hate speech and the system shall immediately send them a warning to stop using hate speech.

4.3 Security Requirements

- 1) Handling with Fraudulents- the admin can remove fraud listing or users if they realize upon inspecting any reports made that the situation was a fraud
- 2) Password requirements- All passwords must be encrypted while registering as a logged in user to ensure the safety of both the users and the platform

4.4 Software Quality Attributes

- 1) Performance- When the user prompts a search, the results are returned within 5 seconds
- 2) Compatibility- The website will be able to run smoothly on most web browsers including chrome, safari, and firefox

5. Other Requirements

5.1 Future changes

- Expand the architectural aesthetics beyond the current 7
- Multi-language support to accommodate for multi-lingual individuals
- Recommendation engine that recommends listings to buyers based on their wishlists

5.2 Unlikely to change

- Database setup (MySQL)- switching away from the setup would require an entire data migration process
- 3 user classes- buyer, seller and admin
- Content to be released in English only for at least the first 2 releases
- Django/ Python backend framework